

Tyler Smith

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EDUCATION

Boston University | Boston, MA — M.S. in Computer Science, Expected Dec. 2027

University of North Carolina at Chapel Hill | Chapel Hill, NC — B.S. in Computer Science & Mathematics, May 2025

EXPERIENCE

Terravela Early-Stage Travel Technology Startup | Software Engineer **Jan 2026 - Present**

- Developed and enhanced a React Native application, implementing backend features, database migrations, UI improvements, and CI/CD workflows across 3 deployment environments.
- Automated the location-based badge creation pipeline, streamlining the addition and deployment of badges across 7+ California cities while replacing 3+ manual database and landmark configuration steps.
- Contributed to OTA deployment and testing workflows with GitHub Actions automating application updates.

Global Career Accelerator | Coding for Web Trainee **June 2024 - Aug 2024**

- Developed and enhanced 50+ web applications with ~5,500 lines of HTML, CSS, and JavaScript, integrating 8+ APIs and earning HTML, CSS, and JavaScript certifications while collaborating with a global development team.

UNC-Chapel Hill | Teaching Assistant For Computer Organization **Jan 2024 - Dec 2024**

- Supported ~200 students across two semesters through 10 weekly office hours, debugging assembly programs and digital circuits while assisting with binary arithmetic, CPU organization, and computer architecture concepts.

UNC-Chapel Hill | Teaching Assistant For Introduction to Programming **August 2022 - Jan 2024**

- Supported 600+ students across 3 semesters through 5–10 weekly office hours, debugging Python code and teaching data structures, recursion, OOP, Git; led a Flask web development workshop for 50+ students during a hackathon.

HIGHLIGHT PROJECTS

Lola's 2.0 ([GitHub](#), [Presentation](#), [Demo](#))

- Built a full-stack food recommendation platform with React, FastAPI, and AWS Nova Pro, cleaning 13,785 products into a 13,024-record dataset with automated missing-value handling.
- Implemented health-based rankings with 20x lower response latency through Pandas preprocessing and developed 50 automated tests across data, APIs, chat services, and frontend components in an Agile team.

Enterprise Backend ([GitHub](#))

- Built a production-grade REST API with TypeScript, Node.js, Express, and PostgreSQL, implementing JWT authentication, role-based access control, and organization management across 7 endpoints.
- Implemented 3-tier API rate limiting for authentication, read, and mutation routes, enforcing 10/15 min, 100/min, and 60/min limits, alongside bcrypt password hashing and protected resource access.
- Developed 9 unit and integration test suites achieving 100% test coverage and configured GitHub Actions CI with PostgreSQL 16, automated coverage, and ESLint validation; maintained comprehensive project documentation.

ML Climate Modelling ([GitHub](#), [Paper](#))

- Developed climate prediction models using Python, TensorFlow, and Scikit-learn across CO₂, temperature, sea-level, rainfall, and extreme-weather datasets; achieved 0.835 R² and 1.889 MAE with Random Forest regression.
- Applied preprocessing, feature engineering, and cross-country model evaluation to assess prediction performance.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C, SQL, MIPS Assembly

Frameworks: React, React Native, Node.js, Express.js, Flask, Bootstrap, Tailwind CSS

Databases & Cloud: PostgreSQL, Supabase, AWS, Docker, Kubernetes

Data & ML: Pandas, NumPy, Scikit-learn, TensorFlow